TUTORIAL 11

Eigenvalues and Diagonalization

MOTIVATION AND GUIDING IDEA

Diagonalization replaces coordinates whose components interact by coordinates in which each component is multiplied by its own scalar. This helps compute powers, understand update processes, and identify preserved or decaying components. Finding eigenvalues is only the first step: enough linearly independent eigenvectors must exist over the field being used.

NOTATION AND CONVENTIONS

- $A \in M_n(\mathbb{F})$ is square; $\lambda \in \mathbb{F}$ is a scalar, and $v \in \mathbb{F}^n \setminus \{0\}$ is an eigenvector when $Av = \lambda v$.
- $E_\lambda(A) = \ker(A - \lambda I_n) = \ker(\lambda I_n - A)$ is the eigenspace. The notations V_λ and $N(\lambda I_n - A)$ used in the notes describe the same space.
- $p_A(t) = \det(tI_n - A)$ is the characteristic polynomial; t is neither a vector nor a matrix parameter. The notations $P_A(x)$ and $\chi_A(t)$ express the same convention with a different variable name.
- $r_a(\lambda)$ is the algebraic multiplicity, and $r_g(\lambda) = \dim E_\lambda(A)$ is the geometric multiplicity.
- $P = (v_1 \ \cdots \ v_n)$ diagonalizes A when its columns form an eigenbasis. If $Av_i = \lambda_i v_i$, then $AP = PD$, with $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$.
- For $f(t) = \sum_{j=0}^{m} c_j t^j$, define $f(A) = \sum_{j=0}^{m} c_j A^j$, where $A^0 = I_n$.

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DEFINITION 11.1

For $\lambda \in \mathbb{F}$ and $A \in M_n(\mathbb{F})$, λ is an eigenvalue exactly when $\ker(A - \lambda I_n) \neq \{0\}$, equivalently $p_A(\lambda) = 0$. The eigenspace contains zero, but zero itself is not an eigenvector. For an eigenvalue, the algebraic multiplicity is the integer $r \geq 1$ such that $p_A(t) = (t - \lambda)^r q(t)$ and $q(\lambda) \neq 0$; the geometric multiplicity satisfies $1 \leq r_g(\lambda) \leq r_a(\lambda)$.

DEFINITION 11.2

A matrix $A \in M_n(\mathbb{F})$ is diagonalizable over $\mathbb{F}$ if there are an invertible $P \in M_n(\mathbb{F})$ and a diagonal $D \in M_n(\mathbb{F})$ with $P^{-1}AP = D$. Equivalently, $\mathbb{F}^n$ has a basis of eigenvectors. The precise criterion is that p_A split into linear factors over $\mathbb{F}$ and that $r_g(\lambda) = r_a(\lambda)$ for every eigenvalue. Splitting is automatic over $\mathbb{C}$; over $\mathbb{R}$, it must be checked.

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NOTE

If $A = PDP^{-1}$, then $f(A) = Pf(D)P^{-1}$ for every polynomial f. In particular, $A^k = PD^kP^{-1}$ for every integer $k \geq 0$; negative powers require invertibility. If $Av = \lambda v$ and $v \neq 0$, then $f(A)v = f(\lambda)v$, but distinct eigenvalues of A can give the same value of f. Diagonalizability of a power or polynomial of A therefore does not imply diagonalizability of A.

NOTE

Sum eigenspace dimensions over distinct eigenvalues. Eigenspaces for distinct eigenvalues form a direct sum. Their dimensions sum to n exactly when there are enough eigenvectors for a basis; a smaller sum means that an eigenbasis is missing.

MOTIVATION AND GUIDING IDEA

By the end of this tutorial, you will be able to identify directions preserved by a linear map, compute eigenspaces and multiplicities, test diagonalizability, and use an eigenbasis to calculate powers and understand an iterative process. We distinguish statements that hold generally from those requiring a particular field or additional hypotheses.

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11.1 Similar Matrices

DEFINITION 11.3

Let $A, B \in M_n(\mathbb{F})$.
We say that A is **similar** to B (and write $A \sim B$) if there is an invertible $P \in M_n(\mathbb{F})$ such that

$$A = P^{-1}BP$$

PROPOSITION 11.1

Let $A, B \in M_n(\mathbb{F})$. They are similar if and only if there exist an n-dimensional vector space V, a linear operator $T : V \to V$, and two ordered bases $\mathcal{E}, \mathcal{F}$ such that

$$A = [T]_{\mathcal{E}}, \qquad B = [T]_{\mathcal{F}}.$$

If P converts coordinates from $\mathcal{E}$ to $\mathcal{F}$, so that $[v]_{\mathcal{F}} = P[v]_{\mathcal{E}}$, then

$$A = P^{-1}BP.$$

Thus similarity describes the same operation in two coordinate systems.

PROPOSITION 11.2

If $A \sim B$, the following quantities and properties are preserved:

1. **Determinant:** $\det(A) = \det(B)$.
2. **Trace:** $\mathrm{tr}(A) = \mathrm{tr}(B)$.
3. **Characteristic polynomial:** $P_A(x) = P_B(x)$.

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4. **Eigenvalues:** A and B have exactly the same eigenvalues, with the same algebraic and geometric multiplicities.
5. **Rank:** $\text{rank}(A) = \text{rank}(B)$.
6. **Diagonalizability:** A is diagonalizable $\iff$ B is diagonalizable.

NOTE

Warning:
Equality of characteristic polynomials (and hence of eigenvalues, determinants, and traces) **does not guarantee** similarity.
A standard counterexample is $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$. Both have characteristic polynomial x^2, but they are not similar: the zero matrix is similar only to itself (prove that the same is true of every scalar matrix).

11.2 Eigenvalues and Eigenvectors

11.2.1 Motivation and Background

MOTIVATION AND GUIDING IDEA

Eigenvalues and eigenvectors help simplify complex systems. Linear operators (and matrices) generally rotate vectors and mix their coordinates. We seek nonzero vectors on which the action is multiplication by a scalar. Such vectors are **eigenvectors** , and the scalar is an **eigenvalue** . Over $\mathbb{R}$, a positive eigenvalue stretches or contracts along the same direction, a negative one also reverses direction, and a zero eigenvalue sends the vector to zero.

Eigenvalues describe models throughout engineering. The following examples show how one algebraic equation appears in different settings:

- **Google's ranking algorithm (PageRank):** In a random browsing model, a transition matrix M sends a probability distribution to the next one. A stationary ranking distribution $\vec{r}$ satisfies
$$M\vec{r} = \vec{r}, \qquad r_i \geq 0, \qquad \sum_i r_i = 1.$$
This is an eigenvalue equation with $\lambda = 1$. The full PageRank model adds a random jump mechanism to address the link structure; the exercise below studies a simplified model.
- **Quantum mechanics:** In a finite-dimensional quantum model, a pure state is a normalized vector and a Hermitian Hamiltonian H represents energy. The eigenvalue equation
$$H\vec{\psi} = E\vec{\psi}$$
identifies states of definite energy and possible energy values. It is an application of the same linear equation; a general description of mixed states requires the additional concept of a density matrix.
- **Systems of differential equations:** If A is diagonalizable and $A = PDP^{-1}$, substituting

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$\vec{y} = P\vec{z}$ into $\vec{y}' = A\vec{y}$ gives

$$P\vec{z}' = AP\vec{z} = PD\vec{z} \quad \Longrightarrow \quad \vec{z}' = D\vec{z}.$$

Each component satisfies $z_i' = \lambda_i z_i$, so $z_i(t) = z_i(0)e^{\lambda_i t}$. For a continuous linear system, all solutions decay to zero if every eigenvalue has negative real part. If an eigenvalue has positive real part, some solutions grow. A zero real part requires further analysis; its sign alone does not imply decay.

- **Classifying critical points in calculus:** For a twice continuously differentiable real function, consider the symmetric Hessian at a critical point. If all its eigenvalues are positive, the point is a strict local minimum; if all are negative, it is a strict local maximum. Positive and negative eigenvalues imply a saddle point. When zero eigenvalues occur without both signs, the second derivative test may be inconclusive.

DEFINITION 11.4

Let V be a vector space over a field $\mathbb{F}$, and let $T : V \to V$ be a linear map (a linear operator). A scalar $\lambda \in \mathbb{F}$ is an **eigenvalue** of T if there is a **nonzero** vector $\vec{v} \in V$ (that is, $\vec{v} \neq \vec{0}_V$) such that

$$T(\vec{v}) = \lambda\vec{v}$$

This is the **eigenvalue equation** . The vector $\vec{v}$ is an **eigenvector** of T associated with λ.

DEFINITION 11.5

Let $A \in \mathbb{F}^{n\times n}$ be a square matrix of size n.
A scalar $\lambda \in \mathbb{F}$ is an **eigenvalue** of A if there is a **nonzero** column vector $\vec{x} \in \mathbb{F}^n$ (that is, $\vec{x} \neq \vec{0}$) such that

$$A\vec{x} = \lambda\vec{x}$$

This is the **eigenvalue equation** . The vector $\vec{x}$ is an **eigenvector** of A associated with λ.

NOTE

Over $\mathbb{R}$, the line span$\{\vec{x}\}$ is invariant. If $\lambda < 0$, direction is reversed; if $\lambda = 0$, the image is the zero vector. The algebraic concept also applies over $\mathbb{C}$, but a complex eigenvalue cannot always be interpreted as stretching a real line.

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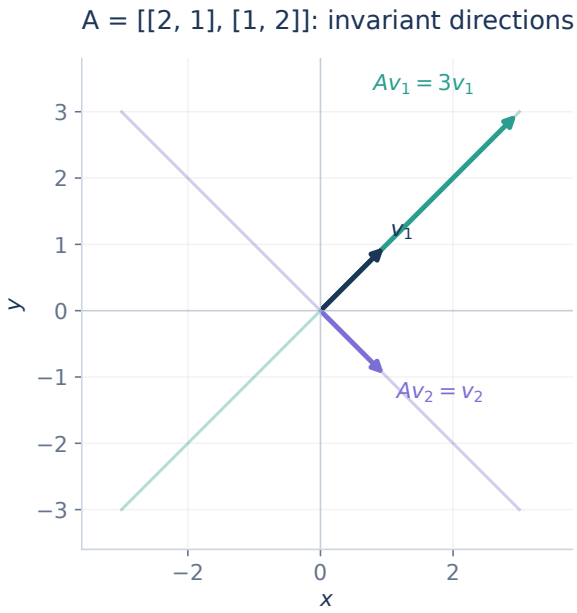


Figure 11.1. Vectors in eigendirections remain on the lines they span. A general vector may change direction because its components are stretched by different factors.

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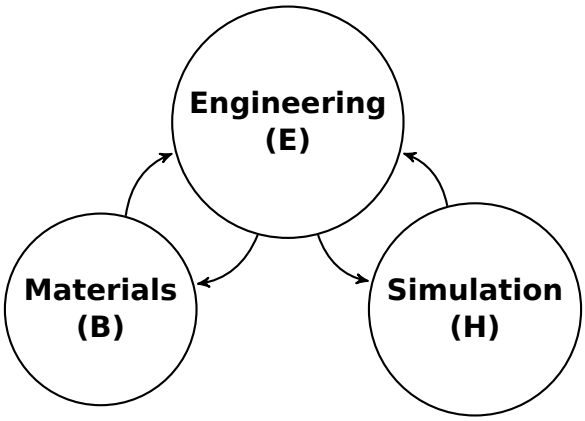
EXERCISE 11.1

Consider a hypothetical internal network at Tel Aviv University consisting of three websites:

- **The central site:** the Faculty of Engineering website (denoted E).
- **Resource site 1:** the materials repository (denoted B).
- **Resource site 2:** the simulation environment (denoted H).

The links are as follows:

- The **materials repository** (B) and **simulation site** (H) link only to the faculty site (E).
- The **faculty site** (E) sends a browser to the materials repository or the simulation environment with equal probability.



Please find a stationary ranking distribution $\vec{r}$ whose components sum to 1. Please compare the faculty site's ranking with those of the two resource sites.

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EXERCISE 11.1 CONTINUED

WORKED SOLUTION

1. Order the sites as 1=Materials (B), 2=Engineering (E), 3=Simulation (H). Construct the transition matrix M:

   - Column B: links only to E, so $M_{2,1} = 1$.
   - Column E: splits equally between B and H, so $M_{1,2} = 0.5$, $M_{3,2} = 0.5$.
   - Column H: links only to E, so $M_{2,3} = 1$.

   The resulting matrix is

   $$M = \begin{pmatrix} 0 & 0.5 & 0 \\ 1 & 0 & 1 \\ 0 & 0.5 & 0 \end{pmatrix}$$

2. Seek the eigenvector $\vec{r} = (r_B, r_E, r_H)^T$ for $\lambda = 1$ satisfying $M\vec{r} = \vec{r}$:

   $$\begin{cases} 0.5r_E = r_B \\ r_B + r_H = r_E \\ 0.5r_E = r_H \end{cases}$$

   The first and third equations immediately give $r_B = r_H = 0.5r_E$.

   Normalize so that the rankings sum to 1:

   $$r_B + r_E + r_H = 1 \implies 0.5r_E + r_E + 0.5r_E = 1 \implies 2r_E = 1$$

   Hence $r_E = 0.5$, and therefore $r_B = 0.25$ and $r_H = 0.25$.

   The ranking vector is

   $$\vec{r} = \begin{pmatrix} 0.25 \\ 0.5 \\ 0.25 \end{pmatrix}$$

3. **Conclusion:** The Faculty of Engineering site ($r_E = 0.5$) has **twice** the ranking of either the materials repository ($r_B = 0.25$) or the simulation environment ($r_H = 0.25$).

NOTE

We found a stationary distribution; we did not prove convergence of every iteration. In fact, browsing initially only at the faculty site produces the cycle

$$(0,1,0)^T \longmapsto (1/2,0,1/2)^T \longmapsto (0,1,0)^T.$$

This model illustrates the difference between existence of a normalized eigenvector and convergence of a process.

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EXERCISE 11.2

Consider the vector space V of all infinite sequences of real numbers. A vector in this space is an infinite sequence

$$\vec{v} = (x_1, x_2, x_3, x_4, \dots)$$

Define the linear operator $T : V \to V$ by multiplying each entry by its position (its index):

$$T(x_1, x_2, x_3, \dots) = (1 \cdot x_1,\ 2 \cdot x_2,\ 3 \cdot x_3,\ 4 \cdot x_4, \dots)$$

For example, if $\vec{v} = (1, 1, 1, \dots)$, then $T(\vec{v}) = (1, 2, 3, \dots)$.
Please find **all** eigenvalues and **all** eigenvectors of T.

WORKED SOLUTION

1. Write $T(\vec{v}) = \lambda\vec{v}$ explicitly for a general vector $\vec{v} = (x_1, x_2, \dots)$:

$$(1x_1,\ 2x_2,\ 3x_3,\ \dots,\ nx_n, \dots) = \lambda \cdot (x_1,\ x_2,\ x_3,\ \dots,\ x_n, \dots)$$

2. Equality of vectors means equality in every coordinate. Compare the nth entries:

$$n \cdot x_n = \lambda \cdot x_n$$

   Rearrange:

$$(n - \lambda) \cdot x_n = 0$$

3. This equation must hold for **every** index $n = 1, 2, 3, \dots$. Suppose λ is an eigenvalue. At every position n with $n \neq \lambda$, the factor in parentheses is nonzero, so necessarily $x_n = 0$.

4. Thus a nonzero vector (with at least one nonzero entry) can occur only if λ is a positive integer.

   Check $\lambda = 1$: The equation is $(n-1)x_n = 0$. For $n = 1$, it becomes $0 \cdot x_1 = 0$ (every x_1 is allowed). For $n \neq 1$, necessarily $x_n = 0$. One eigenvector is $\vec{e}_1 = (1, 0, 0, 0, \dots)$.

   Check $\lambda = 2$: The equation is $(n-2)x_n = 0$. For $n = 2$, it becomes $0 \cdot x_2 = 0$ (every x_2 is allowed). For $n \neq 2$, necessarily $x_n = 0$. One eigenvector is $\vec{e}_2 = (0, 1, 0, 0, \dots)$.

5. **Conclusion:** There are infinitely many eigenvalues: all the positive integers,

$$\lambda_n = n \quad (n = 1, 2, 3, \dots)$$

   For each eigenvalue n, all eigenvectors are $c\vec{e}_n$, where $c \in \mathbb{R} \setminus \{0\}$ and $\vec{e}_n$ is the unit sequence in position n:

$$\vec{e}_n = (0, 0, \dots, \underbrace{1}_{n\text{-th place}}, \dots, 0)$$

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NOTE

The exercise concerns all sequences, with no summability or norm assumption. The unit sequences $\{e_n\}$ are not an algebraic basis of this space: finite linear combinations give only finitely supported sequences, not, for example, $(1, 1, 1, \ldots)$. An algebraically similar model can use indices $n = 0, 1, 2, \ldots$ and the rule $Ne_n = ne_n$. This resembles a number operator. The exercise alone does not imply that all physical energies are discrete or that every physical operator acts on the space of all sequences.

EXERCISE 11.3

Let $n \geq 2$, let $V = M_n(\mathbb{C})$ be the space of square matrices, and let $p, q \in \mathbb{C}$ with $q \neq 0$. Define the linear operator $T : V \to V$ by

$$T(A) = pA + qA^T$$

Please find the eigenvalues and eigenvectors of T directly from the eigenvalue equation.

WORKED SOLUTION

Let $A \neq 0$ be an eigenvector of T with eigenvalue λ. Then

$$pA + qA^T = \lambda A \tag{11.1}$$

Isolate the transpose term:

$$qA^T = (\lambda - p)A$$

Divide by q (which is nonzero) to express A^T in terms of A:

$$A^T = \frac{\lambda - p}{q} A \tag{11.2}$$

Transpose both sides of (11.2):

$$(A^T)^T = \left(\frac{\lambda - p}{q} A\right)^T$$

Using the properties of transposition,

$$A = \frac{\lambda - p}{q} A^T$$

Substitute the expression for A^T from (11.2) into the last equation:

$$A = \frac{\lambda - p}{q} \cdot \underbrace{\left(\frac{\lambda - p}{q} A\right)}_{A^T}$$

$$A = \left(\frac{\lambda - p}{q}\right)^2 A$$

Rearrange and factor out A on the right:

$$\left[1 - \left(\frac{\lambda - p}{q}\right)^2\right] A = 0$$

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EXERCISE 11.3 CONTINUED

Since $A \neq 0$ (an eigenvector), the scalar in brackets must vanish, so $\left(\frac{\lambda-p}{q}\right)^2 = 1$. There are two cases:

1. **First case:** $\frac{\lambda-p}{q} = 1$. Solving for λ gives the first eigenvalue:

$$\lambda_1 = p + q$$

To identify its eigenvectors, substitute $\frac{\lambda-p}{q} = 1$ directly into (11.2):

$$A^T = 1 \cdot A \implies A^T = A$$

Thus the eigenvectors for λ_1 are precisely the nonzero symmetric matrices.

2. **Second case:** $\frac{\lambda-p}{q} = -1$. Solving for λ gives the second eigenvalue:

$$\lambda_2 = p - q$$

Substitute $\frac{\lambda-p}{q} = -1$ into (11.2):

$$A^T = -1 \cdot A \implies A^T = -A$$

Thus the eigenvectors for λ_2 are precisely the nonzero skew-symmetric matrices.

11.3 Eigenspaces, the Characteristic Polynomial, and Multiplicities

DEFINITION 11.6

Definition: the eigenspace V_λ

Let λ be an eigenvalue. The eigenspace V_λ is the set of all eigenvectors for λ together with the zero vector. The zero vector belongs to this subspace but is not an eigenvector. It follows directly from the eigenvalue equation:

1. **For a linear operator T:** Starting from $T(\vec{v}) = \lambda\vec{v}$, rearrange:

$$T(\vec{v}) - \lambda\vec{v} = \vec{0} \implies T(\vec{v}) - \lambda I_V(\vec{v}) = \vec{0}$$

Factor:

$$(T - \lambda I_V)\vec{v} = \vec{0}$$

Thus the eigenspace is the kernel of the operator

$$V_\lambda = \ker(T - \lambda I_V)$$

2. **For a matrix A:** Starting from $A\vec{v} = \lambda\vec{v}$, rearrange:

$$\lambda\vec{v} - A\vec{v} = \vec{0} \implies \lambda I_n\vec{v} - A\vec{v} = \vec{0}$$

Factor out $\vec{v}$ on the right:

$$(\lambda I_n - A)\vec{v} = \vec{0}$$

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Thus the eigenspace is the nullspace of the matrix:

$$V_\lambda = N(\lambda I_n - A)$$

DEFINITION 11.7

Definition: characteristic polynomial and multiplicities

1. **Characteristic polynomial $P_A(x)$:** It is defined by
$$P_A(x) = \det(xI_n - A)$$
2. **Algebraic multiplicity $r_a(\lambda)$:** The multiplicity (exponent) of λ as a root of the characteristic polynomial $P_A(x)$.
3. **Geometric multiplicity $r_g(\lambda)$:** The dimension of the eigenspace for λ:
$$r_g(\lambda) = \dim(V_\lambda) = \dim(N(\lambda I_n - A))$$

EXERCISE 11.4

Consider the real matrix

$$A = \begin{pmatrix} -1 & 2 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}.$$

Please find its characteristic polynomial, eigenspaces, and algebraic and geometric multiplicities. Please determine whether A is diagonalizable.

WORKED SOLUTION

The matrix is triangular, so

$$p_A(\lambda) = \det(\lambda I - A) = (\lambda + 1)^2(\lambda - 2)^2.$$

Both eigenvalues $-1, 2$ have algebraic multiplicity 2. For -1, solve $(A + I)v = 0$:

$$\begin{pmatrix} 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix} v = 0.$$

The equations are $2v_2 = 0, 3v_3 = 0, 3v_4 = 0$, so $E_{-1} = \text{span}\{e_1\}$ and its geometric multiplicity is 1. For 2, the first rows of $A - 2I$ give $-3v_1 + 2v_2 = 0, -3v_2 = 0$. Thus $v_1 = v_2 = 0$, and

$$E_2 = \text{span}\{e_3, e_4\}, \qquad \dim E_2 = 2.$$

The sum of the eigenspace dimensions is $1 + 2 = 3 < 4$, so there is no eigenbasis and A is not diagonalizable.

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11.4 Diagonalization and Eigenbases

MOTIVATION AND GUIDING IDEA

Main objective: find a basis of eigenvectors

DEFINITION 11.8

Definition: diagonalizable matrix and diagonalizing basis

Let $A \in M_n(\mathbb{F})$.

1. A is **diagonalizable over** $\mathbb{F}$ if it is similar over $\mathbb{F}$ to a diagonal matrix. That is, there are an invertible matrix P and a diagonal matrix D, both over $\mathbb{F}$, such that

$$D = P^{-1}AP \iff A = PDP^{-1}$$

   In this case, the diagonal entries of D are the eigenvalues of A:

$$D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$$

2. The columns of the diagonalizing matrix P form a basis of **eigenvectors** of A, in the order of the diagonal entries of D. Since P is invertible, its columns form a basis of $\mathbb{F}^n$. This is a **diagonalizing basis** , or **eigenbasis** .

PROPOSITION 11.3

Diagonalization theorem (criterion for diagonalizability)

A matrix $A \in M_n(\mathbb{F})$ is diagonalizable over $\mathbb{F}$ **if and only if** both conditions hold:

1. **Complete splitting:** The characteristic polynomial $P_A(x)$ splits into linear factors over $\mathbb{F}$, meaning that all its roots (counted with multiplicity) lie in $\mathbb{F}$.
2. **Equal multiplicities:** For each eigenvalue λ, the geometric multiplicity equals the algebraic multiplicity:

$$r_g(\lambda) = r_a(\lambda)$$

Useful consequence: If a matrix A of size n has n **distinct** eigenvalues in $\mathbb{F}$, then $r_a = 1$ for each. Consequently $r_g = 1$ (since $1 \le r_g \le r_a$), and A is diagonalizable over $\mathbb{F}$.

NOTE

Why does $1 \le r_g(\lambda) \le r_a(\lambda)$ always hold? Write $r = \dim V_\lambda$. Since λ is an eigenvalue, a nonzero eigenvector exists, so $r \ge 1$. Extend a basis of V_λ to a basis of the full space. In this basis, the matrix has the form

$$\begin{pmatrix} \lambda I_r & C \\ 0 & D \end{pmatrix}.$$

The lower-left block is zero because each of the first r basis vectors maps to a multiple of itself. Thus the characteristic polynomial is $(x-\lambda)^r \det(xI - D)$, and $r_g(\lambda) = r \le r_a(\lambda)$.

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DEFINITION 11.9

Definition: a diagonalizable linear operator

Let $T : V \to V$ be a linear operator on a finite-dimensional vector space V of dimension n over $\mathbb{F}$. The operator T is **diagonalizable over** $\mathbb{F}$ if there is a basis $\mathcal{B}$ of V in which $[T]_{\mathcal{B}}$ is diagonal.
Such a basis necessarily consists of eigenvectors of T and is called a **diagonalizing basis** .

PROPOSITION 11.4

Let $v_1, \ldots, v_k$ be eigenvectors of a matrix A (or an operator T), associated with **distinct** eigenvalues $\lambda_1, \ldots, \lambda_k$.
Then $\{v_1, \ldots, v_k\}$ is **linearly independent** .

MOTIVATION AND GUIDING IDEA

An algorithm for diagonalizing a linear operator
To diagonalize T (or find a diagonalizing basis), proceed as follows:

1. Choose a convenient basis $\mathcal{C}$ of V (often the standard basis).
2. Form its representing matrix $A = [T]_{\mathcal{C}}$.
3. Solve the matrix problem:
   - Compute the characteristic polynomial $P_A(x)$.
   - Find the eigenvalues λ in $\mathbb{F}$.
   - For each λ, find a basis of the eigenspace (the kernel) V_λ.
4. Test diagonalizability: add the geometric multiplicities ($\sum r_g(\lambda)$). If the sum is less than n, the operator is not diagonalizable over $\mathbb{F}$. If it equals n, continue.
5. Recover the diagonalizing basis: The vectors found in step 3 are coordinate vectors (in $\mathbb{F}^n$). Convert them back to vectors of V using the initial basis $\mathcal{C}$. The combined lists form the diagonalizing basis $\mathcal{B}$.
6. The representing matrix in the new basis is

$$[T]_{\mathcal{B}} = D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$$

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11.4.1 Exercises on Diagonalization

EXAMPLE 11.1

A polynomial that splits into linear factors over $\mathbb{R}$: $p(x) = x(x-1)(x-2)$.

EXAMPLE 11.2

A polynomial that **does not** split into linear factors over $\mathbb{R}$: $q(x) = x^2 + 1$.
Observe that $q(x)$ **does** split into linear factors over $\mathbb{C}$: $q(x) = (x-i)(x+i)$

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EXERCISE 11.5

For each matrix, please determine whether it is diagonalizable over $\mathbb{R}$ and over $\mathbb{C}$. Whenever it is diagonalizable, please give suitable P, D over the relevant field:

$$A_1 = \begin{pmatrix} 4 & 2 \\ 0 & 4 \end{pmatrix}, \quad A_2 = \begin{pmatrix} -1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \quad A_3 = \begin{pmatrix} 0 & -9 \\ 1 & 0 \end{pmatrix}.$$

WORKED SOLUTION

1. For A_1, the characteristic polynomial is $(\lambda - 4)^2$. The equation $(A_1 - 4I)v = 0$ forces $2v_2 = 0$, so the eigenspace is $\text{span}\{(1,0)^T\}$ and has dimension 1. Its algebraic multiplicity 2 exceeds its geometric multiplicity, so it is not diagonalizable over either field.

2. A_2 has distinct eigenvalues $-1, 1, 3$. For -1, choose e_1. For 1, the equation $-2v_1 + 2v_2 = 0$ gives $(1,1,0)^T$. For 3, choose e_3. Hence

$$P = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad D = \text{diag}(-1, 1, 3).$$

   The columns are linearly independent, and their images give $A_2P = PD$. This is a diagonalization over $\mathbb{R}$, and hence also over $\mathbb{C}$.

3. The characteristic polynomial of A_3 is $\lambda^2 + 9$. It has no real root, so there is no real diagonalization. Over $\mathbb{C}$, the eigenvalues are $3i, -3i$, with respective eigenvectors $(3i, 1)^T, (-3i, 1)^T$:

$$A_3(\lambda, 1)^T = (-9, \lambda)^T = \lambda(\lambda, 1)^T \quad \text{when } \lambda^2 = -9.$$

   Therefore

$$P_{\mathbb{C}} = \begin{pmatrix} 3i & -3i \\ 1 & 1 \end{pmatrix}, \quad D_{\mathbb{C}} = \text{diag}(3i, -3i).$$

   Since $\det P_{\mathbb{C}} = 6i \neq 0$, this is a valid complex diagonalization.

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EXERCISE 11.6

As in the earlier transpose exercise, assume $n \geq 2$ and consider $T : M_n(\mathbb{R}) \to M_n(\mathbb{R})$ defined by

$$T(A) = pA + qA^T$$

where $p, q \in \mathbb{R}$ and $q \neq 0$.
Please find the characteristic polynomial $P_T(x)$ and the algebraic and geometric multiplicities (r_a, r_g) of each eigenvalue found earlier.

WORKED SOLUTION

Previously we decomposed the full space $V = M_n(\mathbb{R})$ into a direct sum of eigenspaces:

$$V = V_{p+q} \oplus V_{p-q} = S \oplus K$$

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EXERCISE 11.6 CONTINUED

Here S is the space of symmetric matrices and K is the space of skew-symmetric matrices.

1. **For the eigenvalue $\lambda_1 = p + q$:** The eigenspace consists of the symmetric matrices:

$$V_{\lambda_1} = S = \{A \in M_n(\mathbb{R}) \mid A^T = A\}$$

   Its geometric multiplicity is its dimension (the number of entries on or above the diagonal):

$$r_g(p+q) = \dim(S) = \frac{n(n+1)}{2}$$

2. **For the eigenvalue $\lambda_2 = p - q$:** The eigenspace consists of the skew-symmetric matrices:

$$V_{\lambda_2} = K = \{A \in M_n(\mathbb{R}) \mid A^T = -A\}$$

   Its geometric multiplicity is its dimension (the number of entries strictly above the diagonal, since diagonal entries must be zero):

$$r_g(p-q) = \dim(K) = \frac{n(n-1)}{2}$$

3. **The characteristic polynomial $P_T(x)$:** The geometric multiplicities sum to the dimension of the full space (n^2), so the operator is diagonalizable. Thus each algebraic multiplicity equals its geometric multiplicity ($r_a = r_g$).

   The characteristic polynomial is the product of the linear factors, each raised to its algebraic multiplicity:

$$P_T(x) = (x - \lambda_1)^{r_a(\lambda_1)} \cdot (x - \lambda_2)^{r_a(\lambda_2)}$$

   Substitute the values found above:

$$P_T(x) = \Big(x - (p+q)\Big)^{\frac{n(n+1)}{2}} \cdot \Big(x - (p-q)\Big)^{\frac{n(n-1)}{2}}$$

NOTE

The dimension of a solution subspace can be found by counting independent homogeneous linear constraints:

$$\dim(\text{Subspace}) = \dim(\text{Ambient space}) - (\text{number of independent constraints})$$

Here the ambient space is $M_n(\mathbb{R})$, of dimension n^2.

1. **For symmetric matrices (S):** The condition is $A_{ij} = A_{ji}$. This is a constraint only for $i < j$ (on the diagonal $A_{ii} = A_{ii}$ is an identity, and the equations below the diagonal repeat those above it). There are $\frac{n(n-1)}{2}$ independent constraints, one per entry strictly above the diagonal. Thus the dimension is

$$n^2 - \frac{n(n-1)}{2} = \frac{2n^2 - n^2 + n}{2} = \frac{n(n+1)}{2}$$

2. **For skew-symmetric matrices (K):** The condition is $A_{ji} = -A_{ij}$. There are two types of constraints:

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- For $i < j$, there are $\frac{n(n-1)}{2}$ constraints (as in the symmetric case).
- For $i = j$ (on the diagonal), $A_{ii} = -A_{ii}$ implies $A_{ii} = 0$. There are n such constraints.

The total number of independent constraints is $\frac{n(n-1)}{2} + n$. Thus the dimension is

$$n^2 - \left(\frac{n(n-1)}{2} + n\right) = \frac{n(n-1)}{2}$$

NOTE

A useful check: the sum of the exponents (degrees) in the characteristic polynomial must equal the dimension of the space on which the operator acts.

$$\frac{n(n+1)}{2} + \frac{n(n-1)}{2} = \frac{n^2+n+n^2-n}{2} = \frac{2n^2}{2} = n^2$$

This is indeed the dimension of $M_n(\mathbb{R})$.

EXERCISE 11.7

Let $n \in \mathbb{N}$, and consider a general upper triangular matrix $A \in M_n(\mathbb{R})$:

$$A = \begin{pmatrix} \lambda_1 & * & \cdots & * \\ 0 & \lambda_2 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}$$

Its diagonal entries $\lambda_1, \ldots, \lambda_n$ are **pairwise distinct** (that is, $\lambda_i \neq \lambda_j$ whenever $i \neq j$).
Define the polynomial $P(x) = 2026x^{2025} - 58x^{58}$.
Please prove that there **exists** a matrix $X \in M_n(\mathbb{R})$ satisfying

$$P(X) = A$$

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WORKED SOLUTION

Solution strategy: Although A has unspecified entries (asterisks) above the diagonal, the essential fact is that its diagonal entries are **distinct**. A matrix of size n with n distinct eigenvalues is diagonalizable. We can therefore work with the simpler diagonal matrix, solve the resulting scalar problem using calculus, and return to the original coordinates.

1. The characteristic polynomial of an upper triangular matrix is the product of its diagonal factors:
$$P_A(x) = (x - \lambda_1)(x - \lambda_2)\cdots(x - \lambda_n)$$
Its eigenvalues are therefore exactly $\lambda_1, \ldots, \lambda_n$. They are distinct, so A has n linearly independent eigenvectors. Thus A is diagonalizable: there are an invertible M and a diagonal matrix $D = \text{diag}(\lambda_1, \ldots, \lambda_n)$ such that
$$A = MDM^{-1}$$

2. Seek a solution of the form $X = MEM^{-1}$, where $E = \text{diag}(c_1, \ldots, c_n)$ is a diagonal

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EXERCISE 11.7 CONTINUED

matrix to be constructed. The requirement $P(X) = A$ becomes the following equation for diagonal matrices:

$$P(E) = D$$

Since both are diagonal, this is equivalent to n separate scalar equations:

$$P(c_i) = \lambda_i \quad \text{for every } i = 1, \ldots, n$$

3. Consider the real function $f(t) = 2026t^{2025} - 58t^{58}$.
   - It is continuous.
   - Its leading term has odd degree and positive coefficient, so $\lim_{t\to\pm\infty} f(t) = \pm\infty$.

   The intermediate value theorem implies that f maps onto $\mathbb{R}$. Thus for each eigenvalue λ_i, regardless of the unspecified entries in A, there exists $c_i \in \mathbb{R}$ such that $f(c_i) = \lambda_i$.

4. **Construct the solution:** Choose these c_i as the diagonal entries of E. Define

$$X = MEM^{-1}$$

and verify:

$$\begin{aligned} P(X) &= MP(E)M^{-1} = M\,\mathrm{diag}(P(c_1), \ldots, P(c_n))M^{-1} \\ &= M\,\mathrm{diag}(\lambda_1, \ldots, \lambda_n)M^{-1} = MDM^{-1} = A \end{aligned}$$

This proves existence of the required matrix.

11.5 Diagonalization, Powers, and Parameters

EXERCISE 11.8

Let $A \in M_{4\times 4}(\mathbb{R})$ have 1 on the superdiagonal and 0 elsewhere (that is, $A_{12} = A_{23} = A_{34} = 1$, with all other entries zero).

1. Please show that A is nilpotent, meaning that some positive power is zero, and please find its characteristic polynomial.
2. Please find $\dim\ker(A^2)$ and $\dim\ker(A)$.
3. Please prove that A is not diagonalizable.

WORKED SOLUTION

1. The matrix is upper triangular with zero diagonal, so $P_A(x) = x^4$. By the Cayley-Hamilton theorem, $A^4 = 0$, and hence A is nilpotent.
2. For A, there is one free variable (the first column is zero and the remaining columns are linearly independent), so $\dim\ker(A) = 1$. In A^2, the entries equal to 1 lie on the second superdiagonal. There are two zero columns and two linearly independent nonzero columns, giving two free variables: $\dim\ker(A^2) = 2$.
3. The only eigenvalue is $\lambda = 0$, of algebraic multiplicity $r_a = 4$. The previous part gives

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EXERCISE 11.8 CONTINUED

geometric multiplicity $r_g = \dim\ker(A) = 1$. Since $1 \neq 4$, the matrix is not diagonalizable.

EXERCISE 11.9

Let $A \in M_{n\times n}(\mathbb{F})$, where $\mathbb{F}$ has characteristic different from 2 (so $1_\mathbb{F} + 1_\mathbb{F} \neq 0_\mathbb{F}$, and $2^{-1} = \frac{1}{2}$ exists). Suppose $A^2 = I$.

1. Please prove that every eigenvalue of A lies in $\{1, -1\}$.
2. Please prove that A is diagonalizable. (Hint: express every vector as a sum of vectors in the two eigenspaces.)
3. In this part, assume additionally that $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. Given $\mathrm{tr}(A) = t$, please determine $\dim\ker(A - I)$ and $\dim\ker(A + I)$ in terms of n and t.

WORKED SOLUTION

1. Let $\lambda \in \mathbb{F}$ be an eigenvalue. By definition, there is a nonzero vector $v \in \mathbb{F}^n$, $v \neq 0$, with

$$Av = \lambda v$$

Multiply the equality on the left by A:

$$A(Av) = A(\lambda v)$$

Use associativity on the left and take the scalar outside on the right (linearity):

$$A^2 v = \lambda(Av)$$

Substitute $A^2 = I$ on the left and $Av = \lambda v$ on the right:

$$Iv = \lambda(\lambda v)$$

$$v = \lambda^2 v$$

Move all terms to one side:

$$\lambda^2 v - v = 0 \implies (\lambda^2 - 1)v = 0$$

Since $v \neq 0$ (an eigenvector is nonzero), its scalar coefficient must vanish:

$$\lambda^2 - 1 = 0 \implies \lambda^2 = 1$$

The solutions in the field are $\lambda = 1$ or $\lambda = -1$.

2. To prove diagonalizability, it suffices to find a basis of eigenvectors for the full space. Equivalently, the eigenspaces must span the space: $V = V_1 + V_{-1}$. For any $v \in \mathbb{F}^n$, use the identity below (division by 2 is permitted because the characteristic is not 2):

$$v = \frac{1}{2}(v + Av) + \frac{1}{2}(v - Av)$$

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EXERCISE 11.9 CONTINUED

Denote the two summands by

$$u_1 = \frac{1}{2}(v + Av), \quad u_{-1} = \frac{1}{2}(v - Av)$$

Then $v = u_1 + u_{-1}$. Each summand lies in its corresponding eigenspace; a summand may be zero, in which case it is not itself an eigenvector: **Check** u_1**:** Apply A:

$$A(u_1) = A\left(\frac{1}{2}(v + Av)\right) = \frac{1}{2}(Av + A^2v)$$

Substitute $A^2 = I$:

$$= \frac{1}{2}(Av + v) = u_1$$

Thus $u_1 \in \ker(A - I)$ (and if $u_1 \neq 0$, it is an eigenvector for 1). **Check** u_{-1}**:** Apply A:

$$A(u_{-1}) = A\left(\frac{1}{2}(v - Av)\right) = \frac{1}{2}(Av - A^2v)$$

Substitute $A^2 = I$:

$$= \frac{1}{2}(Av - v) = -\frac{1}{2}(v - Av) = -u_{-1}$$

Thus $u_{-1} \in \ker(A + I)$ (and if $u_{-1} \neq 0$, it is an eigenvector for -1). **Conclusion:** Every vector lies in $\ker(A-I)+\ker(A+I)$. Their intersection is trivial: if $Az = z$ and $Az = -z$, then $2z = 0$, so $z = 0$. Consequently, this direct sum is the entire space. Concatenating bases of the two kernels gives an eigenbasis, so A is diagonalizable.

3. The matrix A is diagonalizable, so it is similar to a diagonal matrix D whose diagonal entries are 1 and -1. Write $k = \dim\ker(A - I)$ (the geometric multiplicity of 1, equal to its number of occurrences on the diagonal). $m = \dim\ker(A + I)$ (the geometric multiplicity of -1, equal to its number of occurrences on the diagonal).

   Two equations are available: 1. The matrix has size n, so the multiplicities sum to n:

$$k + m = n$$

   2. Trace is preserved under similarity, so it equals the sum of the diagonal entries of D:

$$k \cdot (1) + m \cdot (-1) = t \implies k - m = t$$

   This is a system of two equations in the two unknowns k, m. Add the equations:

$$(k + m) + (k - m) = n + t \implies 2k = n + t \implies k = \frac{n + t}{2}$$

   Subtract the second equation from the first:

$$(k + m) - (k - m) = n - t \implies 2m = n - t \implies m = \frac{n - t}{2}$$

   Answer:

$$\dim\ker(A - I) = \frac{n + t}{2}, \quad \dim\ker(A + I) = \frac{n - t}{2}.$$

   The field assumption in this part is essential for interpreting the trace as the difference of integer dimensions. In positive characteristic, the trace determines only their difference

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EXERCISE 11.9 CONTINUED

as a field element, which need not determine the integer dimensions.

EXERCISE 11.10

For $t \in \mathbb{R}$, consider

$$A(t) = \begin{pmatrix} 2 & t & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{pmatrix}.$$

Please find the eigenvalues and their algebraic multiplicities, please compute $\dim \ker(A(t) - 2I)$, and please determine when the matrix is diagonalizable. Please prove a formula for $A(t)^n$ for every $n \geq 1$.

WORKED SOLUTION

Triangularity gives $p_A(\lambda) = (\lambda - 2)^2(\lambda - 5)$. Thus 2 has algebraic multiplicity 2, while 5 has algebraic multiplicity 1. Compute

$$A(t) - 2I = \begin{pmatrix} 0 & t & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

The kernel satisfies $tv_2 = 0$, $3v_3 = 0$. If $t \neq 0$, only v_1 is free, giving dimension 1. If $t = 0$, v_2 is also free, giving dimension 2. The eigenvalue 5 has algebraic multiplicity 1, so its geometric multiplicity is 1. The polynomial splits over $\mathbb{R}$, so $A(t)$ is diagonalizable exactly when $t = 0$. We claim that

$$A(t)^n = \begin{pmatrix} 2^n & n2^{n-1}t & 0 \\ 0 & 2^n & 0 \\ 0 & 0 & 5^n \end{pmatrix} \qquad (n \geq 1).$$

At $n = 1$ this is $A(t)$. Assuming the formula for n, right multiplication by $A(t)$ gives diagonal entries $2^{n+1}, 2^{n+1}, 5^{n+1}$. The $(1, 2)$ entry is the sum of two products:

$$2^n t + (n2^{n-1}t) \cdot 2 = (n + 1)2^n t.$$

All other off-diagonal entries remain zero, giving exactly the formula for $n+1$. This completes the induction. Separately, the zeroth power is $A^0 = I_3$.

EXERCISE 11.11

Let $A \in M_{n\times n}(\mathbb{F})$ be square over $\mathbb{F}$, and let $u, v \in \mathbb{F}^n$ be nonzero column vectors. Define $B = A + uv^T$ (this is a rank-one update).

1. Suppose $x \in \mathbb{F}^n$, $x \neq 0$, satisfies:
   - x is an eigenvector of A with eigenvalue λ (that is, $Ax = \lambda x$).
   - $v^T x = 0$.

   Please prove that x is also an eigenvector of B, and please identify its eigenvalue.
2. Now assume $A = \operatorname{diag}(\alpha_1, \ldots, \alpha_n)$ and $u = v = (1, 1, \ldots, 1)^T$. Please prove that for

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EXERCISE 11.11 CONTINUED

every scalar λ outside $\{\alpha_1, \ldots, \alpha_n\}$,

$$\lambda \text{ is an eigenvalue of } B \iff 1 + \sum_{i=1}^{n} \frac{1}{\alpha_i - \lambda} = 0$$

3. For $n = 3$ and $A = \text{diag}(0, 1, 3)$, please compute the characteristic polynomial of B explicitly (with u, v as in the preceding part).

WORKED SOLUTION

1. Check directly by multiplying B by x:

   $$Bx = (A + uv^T)x$$

   Expand:

   $$= Ax + u(v^T x)$$

   Substitute $Ax = \lambda x$ (because x is an eigenvector of A) and $v^T x = 0$ (by assumption):

   $$= \lambda x + u(0) = \lambda x$$

   Conclusion: We obtained $Bx = \lambda x$. Since $x \neq 0$, it is an eigenvector of B with the same eigenvalue λ.

2. We seek the condition $\det(\lambda I - B) = 0$. Substitute B:

   $$\det(\lambda I - (A + uv^T)) = \det((\lambda I - A) - uv^T)$$

   Write $D = \lambda I - A$. This is diagonal, with entries $\lambda - \alpha_i$. **Why is D invertible?** By assumption, λ differs from every α_i, so all its diagonal entries are nonzero.

   Factor out D inside the determinant:

   $$\det(D - uv^T) = \det(D(I - D^{-1}uv^T))$$

   By multiplicativity ($\det(XY) = \det(X)\det(Y)$),

   $$= \det(D) \cdot \det(I - (D^{-1}u)v^T)$$

   For λ to be an eigenvalue, the full expression must be zero. Since D is invertible, $\det(D) \neq 0$, so the second factor must vanish:

   $$\det(I - (D^{-1}u)v^T) = 0$$

   We use $\det(I + xy^T) = 1 + y^T x$, justifying it even when $y^T x = 0$. If $x = 0$, both sides equal 1. Otherwise, extend x to a basis. In this basis, the image of xy^T lies in $\text{span}\{x\}$ and its first column is $(y^T x, 0, \ldots, 0)^T$. Thus $I + xy^T$ has an upper triangular representing matrix with diagonal $(1 + y^T x, 1, \ldots, 1)$, giving determinant $1 + y^T x$. No nonzero eigenvalue of xy^T needs to be assumed. Here $x = -D^{-1}u$ and $y = v$. Hence the determinant is

   $$1 - v^T(D^{-1}u) = 0$$

   Compute $v^T D^{-1} u$: The diagonal entries of D^{-1} are $\frac{1}{\lambda - \alpha_i}$. Multiplication by the all-ones

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EXERCISE 11.11 CONTINUED

vectors adds them:

$$\sum_{i=1}^{n} \frac{1}{\lambda - \alpha_i} = 1$$

Rearranging gives

$$1 - \sum_{i=1}^{n} \frac{1}{\lambda - \alpha_i} = 0 \implies 1 + \sum_{i=1}^{n} \frac{1}{\alpha_i - \lambda} = 0$$

3. Substitute $\alpha_1 = 0, \alpha_2 = 1, \alpha_3 = 3$ into the formula. The characteristic polynomial is the product of $\det(D)$ and the scalar factor:

$$P_B(\lambda) = \underbrace{\lambda(\lambda-1)(\lambda-3)}_{\det(D)} \cdot \left(1 + \frac{1}{-\lambda} + \frac{1}{1-\lambda} + \frac{1}{3-\lambda}\right)$$

Distribute the polynomial factor over the fractions:

$$= \lambda(\lambda-1)(\lambda-3) - (\lambda-1)(\lambda-3) - \lambda(\lambda-3) - \lambda(\lambda-1)$$

Expand all products:

$$= (\lambda^3 - 4\lambda^2 + 3\lambda) - (\lambda^2 - 4\lambda + 3) - (\lambda^2 - 3\lambda) - (\lambda^2 - \lambda)$$

Collect like terms:

$$= \lambda^3 + (-4 - 1 - 1 - 1)\lambda^2 + (3 + 4 + 3 + 1)\lambda - 3$$

$$= \lambda^3 - 7\lambda^2 + 11\lambda - 3.$$

The rational expression is used only when its denominators are nonzero. After cancellation, it gives a polynomial identity, so the characteristic polynomial formula is valid also at $\lambda = 0, 1, 3$. It can be checked directly from the matrix

$$B = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 4 \end{pmatrix}.$$

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EXERCISE 11.12

Two units exchange information at each step according to

$$\begin{pmatrix} x_{k+1} \\ y_{k+1} \end{pmatrix} = A \begin{pmatrix} x_k \\ y_k \end{pmatrix}, \qquad A = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{pmatrix}.$$

Please find an eigenbasis, an explicit formula for A^k and for the two readings, and their limits as $k \to \infty$.

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EXERCISE 11.12 CONTINUED

WORKED SOLUTION

Compute the action of A on two directions:

$$A\begin{pmatrix}1\\1\end{pmatrix}=\begin{pmatrix}1\\1\end{pmatrix}, \qquad A\begin{pmatrix}1\\-1\end{pmatrix}=\begin{pmatrix}1/2\\-1/2\end{pmatrix}=\frac{1}{2}\begin{pmatrix}1\\-1\end{pmatrix}.$$

These vectors are linearly independent, so

$$P=\begin{pmatrix}1&1\\1&-1\end{pmatrix}, \qquad P^{-1}=\frac{1}{2}\begin{pmatrix}1&1\\1&-1\end{pmatrix}, \qquad D=\operatorname{diag}(1,1/2).$$

For every integer $k\geq 0$,

$$A^k=PD^kP^{-1}=\frac{1}{2}\begin{pmatrix}1+2^{-k}&1-2^{-k}\\1-2^{-k}&1+2^{-k}\end{pmatrix}.$$

Consequently,

$$x_k=\frac{x_0+y_0}{2}+2^{-k}\frac{x_0-y_0}{2}, \qquad y_k=\frac{x_0+y_0}{2}-2^{-k}\frac{x_0-y_0}{2}.$$

The mean is preserved, while the difference $x_k-y_k=2^{-k}(x_0-y_0)$ is halved at each step. Both readings therefore converge to the initial mean $(x_0+y_0)/2$. The eigenvalue 1 describes the preserved component, and $1/2$ describes the decaying component.

EXERCISE 11.13 EXAM QUESTION

Let $\mathbb{C}_3[x]=\{p\in\mathbb{C}[x]:\deg p\leq 3\}$, and let $\gamma\in\mathbb{Z}_{\geq 0}$. Define the linear map $T_\gamma:\mathbb{C}_3[x]\longrightarrow\mathbb{C}_3[x]$ by

$$\begin{aligned}T_\gamma(a_0+a_1x+a_2x^2+a_3x^3)&=a_1-a_0x\\&\quad+(\gamma a_2-a_3)x^2+2a_3x^3.\end{aligned}$$

1. Please find the representing matrix of T_γ in the standard basis $\mathcal{B}=(1,x,x^2,x^3)$.
2. Please determine for which γ the map T_γ is diagonalizable over $\mathbb{C}$.
3. For the smallest γ for which T_γ is diagonalizable, please give the characteristic polynomial.
4. For this same γ, please find a basis in which the representing matrix is diagonal, and please give that diagonal matrix.

WORKED SOLUTION

1. Compute the images of the basis vectors:

$$\begin{aligned}T_\gamma(1)&=-x, &\quad T_\gamma(x)&=1,\\T_\gamma(x^2)&=\gamma x^2, &\quad T_\gamma(x^3)&=-x^2+2x^3.\end{aligned}$$

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EXERCISE 11.13 EXAM QUESTION CONTINUED

The matrix columns are their coordinate vectors in the same order:

$$A_\gamma = [T_\gamma]_{\mathcal{B}} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & \gamma & -1 \\ 0 & 0 & 0 & 2 \end{pmatrix}.$$

2. The matrix consists of two diagonal blocks of size 2. Using the convention $\chi_T(t) = \det(tI - [T])$, its characteristic polynomial is

$$\begin{aligned} \chi_{T_\gamma}(t) &= \det \begin{pmatrix} t & -1 \\ 1 & t \end{pmatrix} \det \begin{pmatrix} t-\gamma & 1 \\ 0 & t-2 \end{pmatrix} \\ &= (t^2+1)(t-\gamma)(t-2). \end{aligned}$$

The eigenvalues are $i, -i, \gamma, 2$. If $\gamma \neq 2$, these are four distinct values, since γ is real. A matrix of size 4 with four distinct eigenvalues is diagonalizable.

If $\gamma = 2$, compute the eigenspace for 2:

$$A_2 - 2I = \begin{pmatrix} -2 & 1 & 0 & 0 \\ -1 & -2 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The first equations give $a_1 = 2a_0$ and $-a_0 - 2a_1 = 0$. Thus $-5a_0 = 0$, so $a_0 = a_1 = 0$ over $\mathbb{C}$. The third equation gives $a_3 = 0$, while a_2 is free. Hence

$$\ker(A_2 - 2I) = \operatorname{span}\{(0,0,1,0)^T\}.$$

The geometric multiplicity of 2 is 1, whereas its algebraic multiplicity is 2. Therefore T_2 is not diagonalizable. In conclusion,

$$T_\gamma \text{ is diagonalizable over } \mathbb{C} \iff \gamma \neq 2.$$

3. The smallest allowed value in $\mathbb{Z}_{\geq 0}$ is $\gamma = 0$, which satisfies the diagonalizability condition. Therefore

$$\chi_{T_0}(t) = t(t-2)(t^2+1).$$

4. Check four eigenpolynomials:

$$\begin{aligned} T_0(1+ix) &= -x + i = i(1+ix), \\ T_0(1-ix) &= -x - i = -i(1-ix), \\ T_0(x^2) &= 0, \\ T_0(x^2 - 2x^3) &= 2x^2 - 4x^3 = 2(x^2 - 2x^3). \end{aligned}$$

These are nonzero eigenpolynomials for distinct eigenvalues, so they are linearly independent. There are $4 = \dim \mathbb{C}_3[x]$ of them, so

$$\mathcal{E} = (1+ix,\ 1-ix,\ x^2,\ x^2 - 2x^3)$$

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EXERCISE 11.13 EXAM QUESTION CONTINUED

is a basis. In this order, the representing matrix is

$$[T_0]_{\mathcal{E}} = \operatorname{diag}(i, -i, 0, 2).$$

EXERCISE 11.14 EXAM QUESTION

Work in $V = \mathbb{R}^4$ with its standard basis $\mathcal{E} = (e_1, e_2, e_3, e_4)$. Define another basis $\mathcal{F} = (f_1, f_2, f_3, f_4)$ by

$$f_1 = e_1, \qquad f_2 = e_1 + e_2, \qquad f_3 = e_2 + e_3, \qquad f_4 = e_3 + e_4.$$

Define $T : V \to V$ by $T(e_j) = f_j$ for $j = 1, 2, 3, 4$, and let $S : V \to V$ have representing matrix in $\mathcal{F}$ given by

$$[S]_{\mathcal{F}} = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -5 \\ 0 & 0 & 5 & 0 \end{pmatrix}.$$

Set $U = T^{-1} \circ S \circ T$.

1. Please compute $[U]_{\mathcal{E}}$.
2. Please determine whether U is diagonalizable over $\mathbb{R}$ and over $\mathbb{C}$. Whenever it is diagonalizable, please find a diagonalizing basis and a diagonal matrix. Over $\mathbb{C}$, consider the extension of U to $\mathbb{C}^4$.
3. Define $W : V \to V$ by $[W]_{\mathcal{E}} = 2I_4$. Please determine whether there is a basis $\mathcal{B}$ with $[U]_{\mathcal{B}} = [W]_{\mathcal{B}}$, and whether there is a basis $\widetilde{\mathcal{B}}$ with $[U]_{\widetilde{\mathcal{B}}} = [W]_{\mathcal{E}}$.

WORKED SOLUTION

1. Set $J = [S]_{\mathcal{F}}$, and let P have columns $f_1, \dots, f_4$ expressed in $\mathcal{E}$. Then

$$P = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad \det P = 1.$$

Thus $[v]_{\mathcal{E}} = P[v]_{\mathcal{F}}$, and $[T]_{\mathcal{E}} = P$ because $T(e_j) = f_j$. The change-of-basis formula gives $[S]_{\mathcal{E}} = PJP^{-1}$, so

$$\begin{aligned} [U]_{\mathcal{E}} &= [T^{-1}]_{\mathcal{E}}[S]_{\mathcal{E}}[T]_{\mathcal{E}} \\ &= P^{-1}(PJP^{-1})P = J. \end{aligned}$$

Therefore

$$[U]_{\mathcal{E}} = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -5 \\ 0 & 0 & 5 & 0 \end{pmatrix}.$$

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EXERCISE 11.14 EXAM QUESTION CONTINUED

2. Using $p_U(t) = \det(tI_4 - J)$, the characteristic polynomial is

$$p_U(t) = \det\begin{pmatrix} t & 1 \\ -1 & t \end{pmatrix} \det\begin{pmatrix} t & 5 \\ -5 & t \end{pmatrix} = (t^2+1)(t^2+25).$$

It has no real roots. Thus there is no nonzero real eigenvector, and in particular U is not diagonalizable over $\mathbb{R}$.

Over $\mathbb{C}$, the four eigenvalues $i, -i, 5i, -5i$ are distinct. Explicit eigenvectors are

$$v_1 = \begin{pmatrix} 1 \\ -i \\ 0 \\ 0 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ i \\ 0 \\ 0 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ -i \end{pmatrix}, \quad v_4 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ i \end{pmatrix}.$$

For example,

$$Jv_1 = \begin{pmatrix} i \\ 1 \\ 0 \\ 0 \end{pmatrix} = iv_1, \qquad Jv_3 = \begin{pmatrix} 0 \\ 0 \\ 5i \\ 5 \end{pmatrix} = 5iv_3,$$

and similarly $Jv_2 = -iv_2$ and $Jv_4 = -5iv_4$. If $Q = (v_1\ v_2\ v_3\ v_4)$, then

$$\det Q = (2i)(2i) = -4 \neq 0.$$

Hence $\mathcal{B}_{\mathbb{C}} = (v_1, v_2, v_3, v_4)$ is a diagonalizing basis, with

$$[U]_{\mathcal{B}_{\mathbb{C}}} = Q^{-1}JQ = \operatorname{diag}(i, -i, 5i, -5i).$$

3. W is the map $v \mapsto 2v$, so in every basis $\mathcal{B}$, $[W]_{\mathcal{B}} = 2I_4$. If $[U]_{\mathcal{B}} = [W]_{\mathcal{B}}$, the two maps would be equal: for every v, the coordinate vectors of $U(v)$ and $W(v)$ would coincide. But $U(e_1) = e_2$, whereas $W(e_1) = 2e_1$. Thus no such basis exists.

A basis of the second type does not exist either. Indeed, if R is the matrix whose columns are the vectors of $\widetilde{\mathcal{B}}$ expressed in $\mathcal{E}$, then

$$R^{-1}JR = 2I_4 \quad \Longrightarrow \quad J = R(2I_4)R^{-1} = 2I_4,$$

contradicting the matrix found above. Both answers remain negative if complex bases are allowed.

EXERCISE 11.15 EXAM QUESTION

Let $n \geq 4$, and let $A \in M_n(\mathbb{R})$ be diagonalizable over $\mathbb{R}$ with exactly two distinct eigenvalues. Suppose

$$\operatorname{rank}(A - I_n) = n - 3, \qquad \operatorname{tr}(A) = 2n - 3, \qquad \det A = 2^{n-3}.$$

1. Please find the two eigenvalues and their algebraic multiplicities.
2. Please give the characteristic polynomial $p_A(t)$.
3. Please compute $\operatorname{tr}(A^k)$ for every $k \in \mathbb{Z}_{\geq 1}$.

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EXERCISE 11.15 EXAM QUESTION CONTINUED

WORKED SOLUTION

1. By rank-nullity,

$$\dim\ker(A - I_n) = n - \operatorname{rank}(A - I_n) = n - (n-3) = 3.$$

Thus 1 is an eigenvalue with a 3-dimensional eigenspace. Since A is diagonalizable, every algebraic multiplicity equals the corresponding geometric multiplicity. Hence the algebraic multiplicity of 1 is exactly 3.

Denote the other eigenvalue by λ. There are exactly two eigenvalues, so λ has algebraic multiplicity $n-3$. The trace is the sum of the eigenvalues counted with multiplicity, so

$$3 \cdot 1 + (n-3)\lambda = 2n - 3.$$

Subtracting 3 gives

$$(n-3)\lambda = 2n - 6 = 2(n-3).$$

Since $n \geq 4$, $n - 3 \neq 0$, and therefore $\lambda = 2$. Thus the eigenvalues are 1 of multiplicity 3 and 2 of multiplicity $n-3$. The given determinant agrees with this result:

$$\det A = 1^3\, 2^{n-3} = 2^{n-3}.$$

The rank, diagonalizability, number of distinct eigenvalues, and trace were sufficient; the determinant provides an additional consistency check.

2. With $p_A(t) = \det(tI_n - A)$,

$$p_A(t) = (t-1)^3(t-2)^{n-3}.$$

Under the alternative convention $\det(A - tI_n)$, multiply this expression by $(-1)^n$.

3. There is an invertible matrix P with $A = PDP^{-1}$, where

$$D = \operatorname{diag}(1, 1, 1, \underbrace{2, \ldots, 2}_{n-3}).$$

Repeated multiplication cancels the intervening factors $P^{-1}P$, giving

$$A^k = PD^kP^{-1}, \qquad D^k = \operatorname{diag}(1, 1, 1, \underbrace{2^k, \ldots, 2^k}_{n-3}).$$

Trace is invariant under similarity, so

$$\operatorname{tr}(A^k) = 3 + (n-3)2^k.$$

For example, at $k = 1$ this gives $3 + 2(n-3) = 2n - 3$, as required by the data.

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Additional practice

EXERCISE 11.16 FOUNDATION

Consider the real matrix

$$A = \begin{pmatrix} 3 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

1. Please find its characteristic polynomial and all eigenspaces.
2. Please give real matrices P, D satisfying $A = PDP^{-1}$.
3. Please find an explicit formula for A^k for every integer $k \geq 0$.

WORKED SOLUTION

1. The block structure gives

$$p_A(t) = (t-1)\det\begin{pmatrix} t-3 & -1 \\ -1 & t-3 \end{pmatrix} = (t-1)((t-3)^2 - 1) = (t-1)(t-2)(t-4).$$

The eigenvalues are distinct. Solving the kernel equations, or direct verification, gives

$$E_4 = \text{span}\{(1,1,0)^T\}, \quad E_2 = \text{span}\{(1,-1,0)^T\}, \quad E_1 = \text{span}\{(0,0,1)^T\}.$$

Each eigenspace has dimension 1, as also follows from its algebraic multiplicity 1.

2. Place the eigenbasis vectors in the columns:

$$P = \begin{pmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad P^{-1} = \begin{pmatrix} 1/2 & 1/2 & 0 \\ 1/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad D = \text{diag}(4,2,1).$$

We have $\det P = -2 \neq 0$, and the eigenvalue equations give $AP = PD$. Thus $A = PDP^{-1}$.

3. Computing PD^kP^{-1} gives

$$A^k = \begin{pmatrix} (4^k+2^k)/2 & (4^k-2^k)/2 & 0 \\ (4^k-2^k)/2 & (4^k+2^k)/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The formula also holds at $k = 0$, giving I_3. For $k \geq 1$, all entries are integers despite the division by 2 in this expression.

EXERCISE 11.17 CHALLENGE

Let $A \in M_n(\mathbb{F})$ have n distinct eigenvalues $\lambda_1, \ldots, \lambda_n$ in $\mathbb{F}$.

1. Please prove that $B \in M_n(\mathbb{F})$ satisfies $AB = BA$ if and only if there is a unique polynomial f of degree at most $n-1$ (including the zero polynomial) with $B = f(A)$.

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EXERCISE 11.17 CHALLENGE CONTINUED

2. Over $\mathbb{R}$, please find such a polynomial for

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad B = \begin{pmatrix} 5 & 3 \\ 3 & 5 \end{pmatrix}.$$

WORKED SOLUTION

1. First, every power of A commutes with A, so every linear combination of its powers commutes with it. Thus $B = f(A)$ implies $AB = BA$. Conversely, assume $AB = BA$. Since A has n distinct eigenvalues in the field, it has an eigenbasis $(v_1, \ldots, v_n)$ with $Av_i = \lambda_i v_i$, and every eigenspace is one-dimensional. We have

$$A(Bv_i) = B(Av_i) = \lambda_i Bv_i.$$

Thus $Bv_i \in E_{\lambda_i}(A) = \operatorname{span}\{v_i\}$, including the possibility $Bv_i = 0$. Hence $Bv_i = \mu_i v_i$ for some $\mu_i \in \mathbb{F}$. Define explicitly

$$f(t) = \sum_{i=1}^{n} \mu_i \prod_{\substack{1 \le j \le n \\ j \ne i}} \frac{t - \lambda_j}{\lambda_i - \lambda_j}.$$

All denominators are nonzero, so this polynomial is defined over $\mathbb{F}$. For $n = 1$, an empty product equals 1. Its degree is at most $n-1$, and at $t = \lambda_i$ only the ith summand remains, giving $f(\lambda_i) = \mu_i$. Therefore

$$f(A)v_i = f(\lambda_i)v_i = \mu_i v_i = Bv_i.$$

The two maps agree on a basis, so $f(A) = B$. If another polynomial g of degree at most $n-1$ satisfies $g(A) = B$, then $(f-g)(\lambda_i) = 0$ for every i. A nonzero polynomial of degree at most $n-1$ cannot have n distinct roots, so $f = g$. Uniqueness requires the degree bound; without it, a multiple of p_A may be added.

2. The vectors $(1,1)^T, (1,-1)^T$ form a common eigenbasis. Their eigenvalues for A are $3, 1$, and for B they are $8, 2$, respectively. Seek $f(t) = at + b$, requiring

$$3a + b = 8, \qquad a + b = 2.$$

Subtracting gives $2a = 6$, so over $\mathbb{R}$ we obtain $a = 3$, $b = -1$. Indeed,

$$B = 3A - I_2, \qquad f(t) = 3t - 1.$$

This numerical example is over $\mathbb{R}$ (or $\mathbb{C}$); the general result in the first part holds over every field in which the stated eigenvalues belong to the field and are distinct.

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EXERCISE 11.18 CHALLENGE

For $n \geq 2$, define $S(x_1, \ldots, x_n) = (x_n, x_1, \ldots, x_{n-1})$ on $\mathbb{C}^n$.

1. Please find an eigenbasis of S and its characteristic polynomial.
2. For a prime p, please define the same shift on $\mathbb{Z}_p^p$ and prove that it is not diagonalizable over $\mathbb{Z}_p$.

WORKED SOLUTION

Let $\omega = e^{2\pi i/n}$ and define

$$v_k = (1, \omega^{-k}, \omega^{-2k}, \ldots, \omega^{-(n-1)k})^T, \qquad 0 \leq k < n.$$

The first coordinate of Sv_k is $\omega^{-(n-1)k} = \omega^k$. Coordinate $j \geq 2$ is $\omega^{-(j-2)k} = \omega^k \omega^{-(j-1)k}$. Hence $Sv_k = \omega^k v_k$. The n eigenvalues are distinct, so these eigenvectors form a basis, and

$$p_S(t) = \prod_{k=0}^{n-1} (t - \omega^k) = t^n - 1.$$

Over $\mathbb{Z}_p$, $S^p = I$. Since S and I commute and $\binom{p}{k}$ is divisible by p for $1 \leq k < p$, the binomial formula gives

$$(S - I)^p = S^p - I = 0.$$

If S were diagonalizable, so would $S - I$ be. A diagonal matrix with a zero positive power has every diagonal entry zero. This would imply $S = I$, contradicting $S(e_1) = e_2 \neq e_1$ for $p \geq 2$.

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EXERCISE 11.19 CHALLENGE

Let $A \in M_n(\mathbb{C})$ and assume A^2 is diagonalizable. Please prove that A is diagonalizable if and only if $\ker A = \ker A^2$. Please give a 2×2 example in which A^2 is diagonalizable and A is not.

WORKED SOLUTION

If A is diagonalizable, an eigenbasis makes it diagonal. A diagonal entry is zero exactly when its square is zero, giving equality of the kernels. Conversely, suppose the kernels agree. Since A^2 is diagonalizable,

$$\mathbb{C}^n = \bigoplus_{\mu} E_\mu(A^2), \qquad E_\mu(A^2) = \{v : A^2 v = \mu v\}.$$

Each summand is preserved by A, since $A^2(Av) = A(A^2 v) = \mu Av$. On $E_0(A^2) = \ker A$, the map A is zero. For $\mu \neq 0$, choose $\lambda \in \mathbb{C}$ with $\lambda^2 = \mu$. For $v \in E_\mu(A^2)$, set

$$v_+ = \tfrac{1}{2}(v + \lambda^{-1} Av), \qquad v_- = \tfrac{1}{2}(v - \lambda^{-1} Av).$$

Then $v = v_+ + v_-$ and direct substitution gives $Av_+ = \lambda v_+$ and $Av_- = -\lambda v_-$. Their eigenspaces intersect trivially because $\lambda \neq 0$. Thus each summand has a basis of eigenvectors for A, and their union is an eigenbasis of the whole space. For a counterexample, $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ has $A^2 = 0$. Its only possible eigenvalue is zero. If A were diagonalizable, it would be zero, contradicting its displayed entry.

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